WEEK 8, BEFORE THE FIRST PASS

Week 8 pre-read.

About 30 minutes. Graded complete or incomplete. Do this before the first pass.

This is a quick look ahead, not a full read. Look at the headings and the figures, answer the questions below in a few words, and move on. You will learn all of it properly in the walkthroughs, and it will make much more sense because you have already seen it once. **This week covers parts of three chapters: Chapter 12 (the autonomic nervous system), Chapter 8 (how hormones work, and the hypothalamus and pituitary), and Chapter 26 (reproductive hormones, pregnancy and birth).** Every section below has a part for each.

Write every answer in your own words

Do not copy from the book, the notes, a website, a classmate or an AI tool, even if you change a few words. Read the source, look away, and then write what you understood. If you cannot say it without looking, that tells you what to go back and study. Your own words, even rough ones, are what I want to see.

Where to look

Read closely, Silverthorn Chapter 12, Efferent Division: Autonomic and Somatic Motor Control: the section on the autonomic division. That covers the two branches, sympathetic and parasympathetic, where they leave the spinal cord, their neurotransmitters and receptors, the adrenal medulla, and how drugs act on them. You read the somatic motor part last week.

Read closely, Silverthorn Chapter 8, Introduction to the Endocrine System: the whole chapter. It covers what makes a chemical a hormone, the three chemical classes of hormones, how hormone release is controlled, the hypothalamus and the pituitary gland, how hormones interact, and what happens when an endocrine gland makes too much or too little.

Then, Silverthorn Chapter 26, Reproduction and Development: read the parts on male reproduction, female reproduction and the menstrual cycle, and pregnancy, labor and delivery, and milk production. Skim the rest of the chapter. The walkthroughs teach what you need from it.

Chapter numbers are for the 9th edition. Other editions number things differently, so go by the chapter title and the topic, and use the search in your eText to find it.

No Silverthorn textbook? Use OpenStax

If you chose not to buy the textbook, use the free OpenStax book, Anatomy and Physiology 2e, and do the same pre-read with it. Read these sections:

- **15.1 Divisions of the Autonomic Nervous System.** The two branches, where they leave the spinal cord, and their neurotransmitters and receptors.
- **15.2 Autonomic Reflexes and Homeostasis.** How the two branches work against each other in the same organ.
- **15.4 Drugs that Affect the Autonomic System.** What blocking or copying a neurotransmitter does.
- **17.1 An Overview of the Endocrine System and 17.2 Hormones.** What hormones are, their chemical classes, how they travel and how their release is controlled.
- **17.3 The Pituitary Gland and Hypothalamus.** The two halves of the pituitary and how the hypothalamus controls each one.
- **27.1 and 27.2, the male and female reproductive systems.** Read the parts on how hormones control sperm production, the ovarian cycle and the uterine cycle.
- **28.4 Maternal Changes During Pregnancy, Labor, and Birth, and 28.6 Lactation.** The hormones of pregnancy, labor and milk.

About the figures. The figures listed on this page are the ones in Silverthorn. For each one, look for a picture of the same thing in those OpenStax sections. If you cannot find a matching picture, write your own list instead: on a sheet of paper, list each picture you did find in those sections and write one sentence about what it shows. Upload that page with your pre-read.

The Find it questions still work with OpenStax. Search the OpenStax sections for the same key words.

Figures to look at closely

These are the most important figures for this week. Find each one in your eText by what it shows, using the search for the words in bold. Read its labels, and look for the thing named after it.

Chapter 12, the autonomic nervous system

- **The two branches side by side.** Find where each branch leaves the spinal cord and where its ganglia sit, close to the cord or close to the organ.
- **The neurotransmitters and receptors.** Find the neurotransmitter at each synapse and ending, and which kind of receptor it binds.
- **The adrenal medulla.** Find why it is called a modified sympathetic ganglion.

Chapter 8, hormones and the pituitary

- **The three classes of hormones.** Compare peptide and steroid hormones: how they are made, stored, carried in the blood and where their receptors are.
- **The hypothalamus and the pituitary.** Find the neurons that end in the posterior pituitary, and the small blood vessels, the portal system, that carry hormones to the anterior pituitary.
- **A feedback loop with three tiers.** Follow a hormone from the hypothalamus to the anterior pituitary to a gland, and find the arrows that feed back.

Chapter 26, reproduction

- **The control of sperm production.** Find which pituitary hormone acts on the Sertoli cells and which acts on the Leydig cells.
- **The menstrual cycle.** Line up the hormone curves, the follicle in the ovary and the lining of the uterus on one time axis. Find the LH surge.
- **Labor.** Find the positive feedback loop that drives contractions.

What this reading should help you do

Read each goal and pick the answer that fits you right now. There is no wrong answer here.

The autonomic nervous system (Chapter 12)

Compare the somatic motor pathway with the two-neuron autonomic pathway.

◯ This is new to me ◯ I have heard of it

◯ I could explain it now

Compare the sympathetic and parasympathetic branches, and predict what each does to heart rate and to digestion.

◯ This is new to me ◯ I have heard of it

◯ I could explain it now

Match each autonomic receptor to its neurotransmitter, a place in the body and the response it makes, and predict what a drug that blocks one would do.

◯ This is new to me ◯ I have heard of it

◯ I could explain it now

Hormones and the pituitary (Chapter 8)

Sort a hormone as a peptide, steroid or amine, and explain how its class decides how it is stored, carried and received.

◯ This is new to me ◯ I have heard of it

◯ I could explain it now

Explain why one hormone can act differently in different tissues, and how hormones can work together or against each other.

◯ This is new to me ◯ I have heard of it

◯ I could explain it now

Trace a hormone axis from the hypothalamus to the pituitary to a gland, with its feedback, and tell a primary disorder from a secondary one.

◯ This is new to me ◯ I have heard of it

◯ I could explain it now

Reproduction (Chapter 26)

Explain how hormones control sperm production, the ovarian and uterine cycles, pregnancy, labor and milk.

◯ This is new to me ◯ I have heard of it

◯ I could explain it now

Find it

Each answer is in the part of the chapter named. A few words is enough.

The autonomic nervous system (Chapter 12)

Which branch has its ganglia close to the spinal cord, and which has its ganglia close to or inside the organ?

In Chapter 12, the section on the autonomic division. Search for sympathetic chain.

What neurotransmitter do all preganglionic autonomic neurons release, and what kind of receptor does it bind on the next neuron?

In Chapter 12, the part on autonomic neurotransmitters. Search for nicotinic.

Which receptor does norepinephrine bind on the heart, and what happens to heart rate when it does?

In Chapter 12, the part on adrenergic receptors. Search for beta 1 or beta-1.

Why is the adrenal medulla called a modified sympathetic ganglion?

In Chapter 12, the part on the adrenal medulla.

Hormones and the pituitary (Chapter 8)

Name the three chemical classes of hormones and give one example of each.

In Chapter 8, the part on the classification of hormones.

Why do steroid hormones travel in the blood bound to carrier proteins, and what does that do to how long they last?

In Chapter 8, the part on steroid hormones. Search for half-life.

How is the anterior pituitary different from the posterior pituitary? Which one is really part of the brain?

In Chapter 8, the part on the pituitary gland.

What is the hypothalamic-hypophyseal portal system, and why does the body need it?

Same part of Chapter 8. Search for portal.

What is the difference between a primary and a secondary endocrine disorder?

In Chapter 8, the part on endocrine pathologies. Search for secondary.

Reproduction (Chapter 26)

What do the Sertoli cells and the Leydig cells each do?

In Chapter 26, the part on male reproduction. Search for Leydig.

What causes the LH surge in the middle of the menstrual cycle, and what does the surge set off?

In Chapter 26, the part on the menstrual cycle. Search for LH surge.

Which hormone keeps the corpus luteum alive in early pregnancy, and what makes it?

In Chapter 26, the part on pregnancy. Search for chorionic.

Words to know

Find each term in the reading. Write what it means in your own words, not the book's. If you are working on paper, add a quick sketch. If you are typing, give one example instead.

Preganglionic neuron

Dual innervation

Muscarinic receptor

Half-life

Down-regulation

Trophic hormone

Long-loop negative feedback

Posterior pituitary

Corpus luteum

Positive feedback

Before you start the walkthroughs

Which of this week's topics looks hardest to you right now, and why?

Write one question you want the walkthroughs to answer.

Turn it in

Save this page as a PDF with the button below and upload it to the Week 8 pre-read assignment in Canvas. If you worked on paper, upload a photo of your pages instead.